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Differentiating Between Depression, Hopelessness, and Psychache in University Undergraduates

Michelle M. DeLisle
Ronald R. Holden
Queen's University, Kingston, Ontario, Canada

The overlap between depression, hopelessness, and psychache constructs was investigated using 587 undergraduates. Analyses indicated three correlated dimensions; among these, psychache accounts for more variance in depression and hopelessness than these latter variables account for in psychache. All constructs demonstrated convergent validity, but psychache was associated with the widest range of suicide criteria. These findings support that psychache is a leading variable associated with suicide risk.

Keywords: *psychache; hopelessness; depression; suicidality*

Suicide is a major public health concern. In the United States, the suicide rate is 11 per 100,000 people, which translates to an average of 85 suicide deaths every day (Moscicki, 1999). The enormous scope of this issue can be further understood by considering the potential impact of suicide attempts in general. For every suicide death, there are up to 10 times more suicide attempts that result in hospitalization (Holley, Fick, & Love, 1998) and between 50 to 100 times more suicide attempts that result in injuries not requiring inpatient admission (Pagliaro, 1995). To respond to calls for broad-based national suicide prevention strategies (Weir, 2001), it is imperative to first understand the key conditions associated with suicide risk. Suicidal behavior is complex and multiply determined; as such, suicide prevention programs can address only potentially modifiable risk factors (Brown, Beck, Steer, & Grisham, 2000). Research over the past 40 years has emphasized the role of psychological variables in suicide risk, and of these, depression, hopelessness, and psychache have gained preeminence. The time is now ripe for developing more integrative models of suicidality. An important step toward this goal lies in understanding the degree of overlap or independence between the three key statistical suicide prediction models.

Depression, Hopelessness, and Psychache

Beck's (1967) cognitive model is the broadest in scope, and it proposes that the primary causes of depression are one's dysfunctional beliefs about the self, the world, and the future—together constituting a depressive triad. Approximately two thirds of individuals, at the time of their first hospitalization for attempting suicide, are diagnosed

Authors' Note: This research was supported in part by grants from the Social Sciences and Humanities Research Council of Canada. Correspondence concerning this article should be addressed to Ronald R. Holden, Department of Psychology, Queen's University, Kingston, Ontario K7L 3N6, Canada; e-mail: holdenr@queensu.ca.

with depression (Langlois & Morrison, 2002), which has long been associated with suicide completion, based on clinical observation data and empirical research (McHugh & Goodell, 1971; Robins, Schmidt, & O'Neil, 1959; Silver, Bohnert, Beck, & Marcus, 1971). However, variables other than depression have since emerged as strong statistical predictors of suicide risk.

In the early process of investigating the nature of the relationship between depression and suicide, researchers discovered that hopelessness could account for the effects of depression on suicide intent among samples of suicide attempters (Minkoff, Bergman, Beck, & Beck, 1973), general inpatients (Wetzel, Margulies, Davis, & Karam, 1980), and college students (Cole, 1988). As well, clinical ratings of hopelessness (Beck, Brown, & Steer, 1989a, 1989b) and self-report ratings of hopelessness (Beck, Brown, Berchick, Stewart, & Steer, 1990; Beck, Steer, Kovacs, & Garrison, 1985) have been found to successfully predict more than 90% of cases involving patients' dying by suicide 5 to 10 years after treatment termination. Taken together, these findings suggest that in a range of populations, suicide is more strongly associated with hopelessness than depression. Abramson, Metalsky, and Alloy (1989) have developed a streamlined model that focuses on the future aspect of Beck's (1967) original cognitive triad. In their model, beliefs need not be distorted or unrealistic but must involve an expectation that negative outcomes will occur or that positive outcomes will not occur, as well as a sense of helplessness about changing the probability of these outcomes. Abramson and colleagues propose that such attributions lead to a subtype of depression, termed *hopelessness depression*, of which suicidality is a symptom—but not of depression in general.

More recently, Shneidman (1993) has proposed that suicide is caused by *psychache*, or what he referred to as the anguish, hurt, angst, or humiliation that leads individuals to seek permanent escape from unbearable levels of psychological pain. Shneidman does not conceptualize suicide as a symptom of a psychiatric disorder but rather as the behavioral outcome that results when individuals have a high level of psychache, a lowered threshold for enduring psychological pain, and a perception of death as the only solution. Evidence continues to accumulate suggesting that psychache or internal perturbations contribute unique explanatory variance to the statistical prediction of suicidality, when controlling for other factors such as hopelessness (Berlim et al., 2003; DeLisle & Holden, 2004; Holden, Kerr, Mendonca, & Velamoor, 1998; Holden & Kroner, 2003; Holden & McLeod, 2000; Holden, Mehta, Cunningham, & McLeod, 2001; Johns & Holden, 1997). Yet, the place of psychache in relation to theories of hopelessness and depression is not currently well articulated.

Scale Measurement Issues

The present study is the first, to our knowledge, to clarify the degree of empirical overlap between depression, hopelessness, and psychache. We selected the Beck Depression Inventory (BDI; Beck & Steer, 1987) and the Beck Hopelessness Scale (BHS; Beck, Weissman, Lester, & Trexler, 1974) to measure depression and hopelessness because of their longstanding use for theory testing by the originators of key cognitive models. Abramson et al. (1989) favored the BHS in particular because its items assess generalized hopelessness and not simply circumscribed pessimism; furthermore, the BHS provides an operational definition of hopelessness that is distinct from the symptoms of hopelessness depression, as proposed by their model. Likewise, the Psychache Scale (Holden et al.,

2001) is the most psychometrically sound measure of psychache available (Davie, 2005; Mills, Green, & Reddon, 2005).

However, the different response formats of the BDI (0–3 scale), BHS (true/false), and Psychache Scale (1–5 scale) are potentially problematic for delineating construct independence. As such, the tests introduce method variance (Wiggins, 1962), which refers to differences in scores owing to the structural aspects of the instruments (e.g., response format). Such aspects can obscure differences in scores attributed to true variance in the constructs being assessed, such as real levels of depression, hopelessness, and psychache. To more clearly determine the impact of method variance on the overlap of the BDI, BHS, and Psychache Scale items, participants in the current study completed the questionnaires in their standard formats, but the data were analyzed in four ways, which presented different amounts of item-level variability—namely, as original items, dichotomized items, parcels of original items, and parcels of dichotomized items. Dichotomization of the BDI and Psychache Scale items was performed using a median split for each item; this method provided item endorsement frequencies that most closely approximated those of the BHS. Individual items, particularly if they are dichotomous, tend to have low reliability, low intercorrelations, and, in factor analysis, low communalities, all of which can lead to difficulties in interpretation; by parceling items, however, a potential solution to such problems emerges (Kishton & Widaman, 1994).

A parcel is the simple sum of several items that compose a scale. In general, several parcels are developed from the items on a scale, with each item being assigned to a parcel only once. A common practice is to ensure that the communalities of each parcel are equal (Meade & Kroustalis, 2006). In the present study, items for each measure were parceled by maximum likelihood extraction of one factor and by distribution of items into three- or four-item parcels with approximately equivalent average factor loadings. Parceling requires that data be normally distributed (Hau & Marsh, 2004). Conclusions based on analyses using normalizing transformations and untransformed data were virtually identical; so, only findings derived from untransformed scales are reported here, for ease of interpretation. Furthermore, only analyses that were consistent across two or more data sets are discussed (i.e., of original, dichotomous, or parceled items) because these findings were perceived to be more robust to the effects of method variance.

Maximum likelihood exploratory factor analysis and confirmatory factor analysis were conducted to determine the dimensionality underlying depression, hopelessness, and psychache combined. The fit was compared between the data and competing nested models—with zero, one, two, and three correlated factors. A good fit for the model with no correlated factors would suggest that depression, hopelessness, and psychache are distinct constructs, whereas a good fit for the models with one, two, and three correlated factors would suggest increasing degrees of overlap.

The potentially asymmetrical nature of any overlap was further examined using canonical correlation analysis with pairs of constructs. For example, Abramson and colleagues' (1989) hopelessness depression model predicts that depression should account for a greater amount of variance in hopelessness, as opposed to the other way around. Finally, structural equation modeling was undertaken to determine the unique contributions of depression, hopelessness, and psychache to a range of suicide criteria—namely, number of attempts, internal perturbation–based reasons for attempting suicide, extrapunitive/manipulative motivations for attempting suicide, suicide motivation, and suicide preparation.

In particular, we propose that depression, hopelessness, and psychache are distinct but related constructs, as demonstrated by

- three dimensions underlying depression, hopelessness, and psychache items;
- a better fit to a model with three correlated factors than to competing models;
- a greater proportion of variance in depression and hopelessness, as accounted for by psychache (as opposed to vice versa), in line with Shneidman's (1985, 1993) view of these constructs as particular components of psychache—specifically, that hopelessness is the common emotion in psychache and that depression is an illness that may or may not be present in suicidal individuals; and
- psychache's being of equal or greater importance than hopelessness and depression in the statistical prediction of suicidal manifestations.

Method

Participants

In sum, 587 volunteer undergraduate participants were recruited from a first-year psychology participant pool (98%), as well as from upper-year classes in psychology (2%). Participants ranged in age from 15 to 45 years (M = 18.72, SD = 2.49), and 78% were women. Thirty-four participants reported having previously attempted suicide, with drugs and poison being the most common method (48.5%), followed by cutting (42.2%) and gases and vapors (3.0%). Suicide attempters reported a mean of 22.63 months since their most recent suicide attempt (SD = 25.48, range = 0.33 to 48.00), a moderate level of suicide intent at the time of that attempt (M = 2.48, SD = 1.46, range = 1 to 5), and a low likelihood of subsequently attempting suicide (M = 1.44, SD = 0.70, range = 1 to 5). No information was collected on race or ethnicity.

Psychological Statistical Predictors

BDI. The BDI (Beck & Steer, 1987) is a 21-item inventory that assesses depression severity in adolescents and adults. Responses are coded on a 4-point scale on which symptoms increase in severity from 0 to 3. Participants are asked to respond on the basis of their experience over the past week, including the day of testing. The BDI has generally demonstrated adequate reliability, with alpha reliability coefficients ranging from .73 to .95 (Beck, Steer, & Garbin, 1988). The concurrent validity of the BDI has also been established in psychiatric and nonpsychiatric samples (Beck, Steer, & Garbin, 1988; Spreen & Strauss, 1991).

BHS. The BHS (Beck et al., 1974) measures the extent of negative expectancies about the future, and it consists of 20 true/false statements. The BHS has been demonstrated to have strong psychometric properties. Alpha reliability coefficients have ranged from .65 to .93 in student, forensic, and psychiatric samples (Beck et al., 1974; Holden, 1986). Correlations of the BHS with clinical ratings of hopelessness have ranged from .66 to .74 (Beck et al., 1974), and BHS scores of 9 and above have accurately predicted death by suicide in outpatients (Beck et al., 1990).

Psychache Scale. The Psychache Scale (Holden et al., 2001) consists of 13 items (e.g., "My soul aches") that measure psychache, or psychological pain. Responses are coded on a 5-point Likert-type scale. The

Psychache Scale has excellent psychometric properties in university undergraduates and prison inmates, with alpha reliability coefficients generally exceeding .90 (Holden et al., 2001; Mills et al., 2005). The Psychache Scale has been demonstrated to distinguish suicide attempters from nonattempters, as well as to statistically predict suicidality when the effects of depression and hopelessness have been controlled (DeLisle & Holden, 2004; Holden et al., 2001; Mills et al., 2005).

Suicide Criteria

Reasons for Attempting Suicide Questionnaire. The Reasons for Attempting Suicide Questionnaire (Holden, Kerr, Mendonca, & Velamoor, 1998) consists of 14 items that assess the motivation for suicide in clinical and nonclinical populations. Responses are coded on a 5-point Likert-type scale ranging from 1 (*completely disagree*) to 5 (*completely agree*). The questionnaire yields two scales that reflect reasons reported by patients for attempting suicide. These include a six-item Internal Perturbation Scale (e.g., "to get relief from a terrible state of mind") and an eight-item Extrapunitive/Manipulative Motivations Scale (e.g., "to frighten someone"). These two factors have been substantiated in suicide attempters (Holden & DeLisle, 2006). Alpha reliability coefficients in university students and attempters have ranged from .71 to .87 for the Internal Perturbations Scale and from .80 to .86 for the Extrapunitive/Manipulative Motivations Scale (Holden & DeLisle, 2006; Holden & McLeod, 2000; Holden et al., 1998; Johns & Holden, 1997).

Beck Scale for Suicide Ideation. The Beck Scale for Suicide Ideation (Beck & Steer, 1993) is a 19-item rating scale that gauges suicidal intent and monitors the quality and quantity of ongoing suicidal ideation. This scale has yielded alpha reliability coefficients of .90 and .87 in inpatient and outpatient ideators (Beck & Steer, 1993), as well as correlations with a clinician-administered version for inpatients and outpatients—respectively, .90 and .94 (Beck, Steer, & Ranieri, 1988). The Beck Scale for Suicide Ideation has two subscales (Beck, Brown, & Steer, 1997): The Motivation subscale taps individuals' ambivalence about living or dying, as well as the frequency and duration of suicidal thoughts; the Preparation subscale refers to a more active stage, one that involves planning the act. Holden and DeLisle (2005) found support for this two-factor structure in a sample of suicide attempters, reporting alpha reliability coefficients of .85 (Motivation subscale) and .73 (Preparation subscale).

Procedure

Before data collection, the research was approved by a university-based general research ethics board. Participants who gave informed consent were asked to complete a questionnaire package consisting of the BDI, BHS, Psychache Scale, the Reasons for Attempting Suicide Questionnaire, and the Beck Scale for Suicide Ideation—in that order. Questionnaires with the least sensitive questions were administered first, to minimize the likelihood of nonresponding owing to concerns about privacy or the consequences of reporting that may be prompted by items about suicide (see Tourangeau & Smith, 1996). Participants also completed a demographic sheet in which they reported their age and gender, their number of previous suicide attempts, their methods used, their intent to kill oneself, and their self-rated likelihood of future suicide attempts. A blank page was attached to allow participants to make additional comments. A debriefing sheet was provided that included

Table 1
Descriptive Statistics and Correlations Between Measures of Suicidality (N = 587)

Scale	Possible Range	Observed Range	Coefficient α	1	2	3	4	5	6	7	8	9	10
1	0–63	0–55	.89	—									
2	0–20	0–20	.85	.68**	—								
3	13–65	13–65	.95	.76**	.62**	—							
4	6–30	6–30	.88	.55**	.44**	.59**	—						
5	8–40	8–33	.86	.34**	.26**	.37**	.60**	—					
6	0–18	0–17	.79	.60**	.51**	.61**	.55**	.30**	—				
7	0–18	0–16	.75	.48**	.36**	.51**	.55**	.24**	.70**	—			
8	0+	0–7	—	.35**	.23**	.33**	.34**	.12**	.53**	.53**	—		
9	1–5	1–5	—	.44**	.54**	.50**	.42*	.15	.55**	.68**	.39*	—	
10	1–5	1–3	—	.27	.23	.56*	.57*	.56*	.68**	.65*	.57*	.33	—
			M	7.83	3.72	21.91	11.01	13.36	1.07	3.10	0.13	2.49	1.44
			SD	6.88	3.69	9.47	5.47	5.96	2.13	2.55	0.65	1.46	0.70

Note: 1 = Beck Depression Inventory; 2 = Beck Hopelessness Scale; 3 = Psychache Scale; 4 = Reasons for Attempting Suicide Questionnaire–Internal Perturbations Scale; 5 = Reasons for Attempting Suicide Questionnaire–Extrapunitive/Manipulative Motivations Scale; 6 = Beck Scale for Suicide Ideation–Motivation; 7 = Beck Scale for Suicide Ideation–Preparation; 8 = number of previous attempts; 9 = suicide intent; 10 = self-reported likelihood of future suicidality. Some descriptive statistics and correlations were based on fewer than 587 participants because of missing data.
*p < .05. **p < .01.

phone numbers of local crisis hotlines. Participants from the introductory psychology participant pool received one credit toward their psychology course research requirement, whereas upper-year psychology students were entered in a draw to win $150 at the end of the term.

Results

Table 1 displays descriptive statistics and intercorrelations between all measures. Observed scores varied across the range of possible values, and all scale coefficient alpha reliabilities exceeded .75. Mean scale scores were generally comparable to those in previous studies employing student samples (Durham, 1982; Flamenbaum & Holden, 2007; Holden et al., 2001; Velting, 1999). The BDI and BHS scores were slightly higher than those for university students in earlier studies by Cole (1988; M_{BDI} = 6.7; M_{BHS} = 2.3) and by Holden et al. (M_{BHS} = 2.7). Although the present BHS scores approached those for male prison inmates found by Holden and Kroner (2003; M = 3.88), inmates' scores for the Reasons for Attempting Suicide Questionnaire (Internal Perturbations Scale and Extrapunitive/Manipulative Motivations Scale) were comparable to those typically reported in general student samples.

Correlations between statistical suicide predictors were moderately high, ranging from .62 (hopelessness and psychache) to .76 (depression and psychache). Psychache and internal perturbations were positively associated but distinct (r = .59). As well, internal reasons for attempting suicide, suicidal motivation, and suicide intent were all more highly correlated with depression, hopelessness, and psychache than were externally based reasons for attempting suicide, suicide preparation, number of previous suicide attempts, and self-reported future likelihood of attempting suicide.

Exploratory Factor Analysis

To determine the number of dimensions underlying BDI, BHS, and Psychache items, four criteria were utilized. First, eigenvalues for item intercorrelations were computed, and those exceeding unity were considered. Because this heuristic tends toward overextraction (Nunnally & Bernstein, 1994), a scree test, Velicer's (1976) minimum average partial criterion, and 95th-percentile values of Horn's (1965) parallel analysis criterion were examined for convergence. Velicer, Eaton, and Fava (2000) recommend the use of these techniques, which were originally developed for principal components analysis, owing to the dearth of equivalent tests for use in factor analysis. For instance, parallel analysis of adjusted correlation matrices with squared multiple correlations on the diagonal tends to indicate more factors than are warranted (Buja & Eyuboglu, 1992). To partly address the issue of different response formats in the BDI, BHS, and Psychache Scale, the criteria for determining dimensionality were applied to the original questionnaire items, to dichotomized items, to parcels of original items, and to parcels of dichotomized items.

Eigenvalue greater than one, minimum average partial criterion, parallel analysis criterion, and scree test failed to converge for analyses of original items (2 to 11 factors) and dichotomized items (2 to 15 factors). The presence of communalities greater than one for these latter-item formats in subsequent analyses suggests interpretability problems. For parcels of original items, the criteria supported both two- and three-factor models. The scree test and parallel analysis criterion converged on two factors (60.38% of the total variance), and two eigenvalues exceeded unity (8.72, 1.55). Minimum average partial criterion supported three factors (66.08% of the total variance). For parcels of dichotomized items, the three-factor model received support. Minimum average partial criterion and parallel analysis criterion converged on three factors (55.47% of the total variance), and three eigenvalues exceeded unity (6.36, 1.79, 1.29). The scree test suggested the presence of two factors composing 47.90% of the total variance.

Based on the previous convergence on two or three dimensions, maximum likelihood extraction of two- and three-factor models was performed using parcels of original items and parcels of dichotomized items (see Table 2). Promax rotation was used to improve interpretability, and loadings less than .30 (9% of the variance) were suppressed. For parcels of original items and parcels of dichotomized items, forcing two factors resulted in less clear differentiation. For original item parcels, psychache was distinct from hopelessness, whereas the majority of depression items overlapped with both constructs. For parcels of dichotomized items, some depression items were associated with hopelessness, whereas other depression items were associated with psychache, with none of the items demonstrating overlap with both. However, when three factors were forced on parcels of original items, depression, hopelessness, and psychache all appeared to be distinct. A similar result for the three-factor model using parcels of dichotomized items emerged, although two of the depression parcels failed to yield loadings greater than .30 on any factor.

Confirmatory Factor Analysis

Because exploratory analyses determined that the three-factor model was more stable

Table 2
Promax Rotated Factor Loadings From Maximum Likelihood Extraction of Two- and Three-Factor Models

	Parcels of Original Items					Parcels of Dichotomous Items				
	Two Factors		Three Factors			Two Factors		Three Factors		
Item	1	2	1	2	3	1	2	1	2	3
D1	.49	—	.55	—	—	.31	—	.41	—	—
D2	.55	—	.59	—	—	—	.32	.48	—	—
D3	.56	—	.49	—	—	—	.33	—	—	—
D4	.40	.36	.90	—	—	.36	—	.85	—	—
D5	.35	.42	.87	—	—	.45	—	—	—	—
D6	.40	.34	.51	—	—	.36	—	.51	—	—
D7	.35	.37	.55	—	—	.46	—	—	.66	—
H1	.69	—	—	.71	—	—	.62	—	.66	—
H2	.67	—	—	.73	—	—	.59	—	.64	—
H3	.61	—	—	.58	—	—	.69	—	.67	—
H4	.68	—	—	.61	—	—	.56	—	.54	—
H5	.61	—	—	.69	—	—	.64	—	.68	—
H6	.67	—	—	.67	—	—	.59	—	.63	—
P1	—	.91	—	—	.90	.86	—	—	—	.86
P2	—	1.01	—	—	.92	.90	—	—	—	.86
P3	—	.91	—	—	.82	.80	—	—	—	.77
P4	—	.97	—	—	.90	.88	—	—	—	.86

Note: Items D1 to D7 refer to parcels of Beck Depression Inventory items. Items H1 to H6 refer to parcels of Beck Hopelessness Scale items. Items P1 to P4 refer to parcels of Psychache Scale items.

when item format varied (i.e., parcels of original versus dichotomized items), the fit of four structural models based on three underlying dimensions was tested using maximum likelihood estimation in AMOS 6.0 (Arbuckle, 2005) to further investigate how depression, hopelessness, and psychache overlap. Of particular interest was whether these three constructs would emerge as being mutually coextensive, whether depression and hopelessness would overlap with each other but remain distinct from psychache, or whether all three constructs would emerge as being independent. An intermediate model—one in which hopelessness was allowed to correlate with depression and psychache—was included to allow all subsequent models to be nested. Research has shown that conclusions based on confirmatory factor analysis test statistics are problematic—namely, when univariate skewness and kurtosis values exceed 2 and 7, respectively (Curran et al., 1996). As such, bootstrapping was applied for undertaking tests of significance. Bootstrapping makes no assumptions about the sample (i.e., normality), because it calculates the characteristics of the sampling distribution empirically—that is, from the data provided. In bootstrapping, the data are treated as a population, and sample sizes are randomly drawn with replacement. With each resampling, the statistic of interest is calculated so that the final distribution of this statistic can be used to generate confidence intervals and to make inferences about population parameters for significance testing (Mooney & Duval, 1993). In the present study, bootstrapping with 10,000 resamples was used to generate confidence intervals around path coefficients, covariances, and factor loadings in the structural model. AMOS 6.0 performs bootstrapping when there are no missing data in any

Table 3
Confirmatory Factor Analyses for Three-Factor Models With Bootstrapping (10,000 Resamples)

Three-Factor Model	*r*	Range of Standardized Factor Loadings			Indices of Model Fit					
		BDI	BHS	PS	χ^2	*df*	RMSEA	CFI	TLI	$\Delta\chi^2$
Parcels of original items										
BDI, BHS, PS	—	.69–.79	.65–.73	.87–.95	1,105.31***	119	.12	.84	.82	—
BDI, BHS	.74**	.69–.78	.62–.73	.87–.95	812.05***	118	.10	.89	.87	293.26***
PS	—									
BDI, BHS	.40**	.65–.74	.64–.73	.87–.95	570.24***	117	.08	.93	.91	241.81***
BHS, PS	.70**									
BDI, PS	—									
BDI, BHS	.74**	.70–.78	.62–.74	.87–.95	330.18***	116	.06	.97	.96	240.06***
BHS, PS	.66**									
BDI, PS	.81**									
Parcels of dichotomous items										
BDI, BHS, PS	—	.52–.74	.65–.73	.78–.91	933.00***	119	.11	.83	.80	—
BDI, BHS	.70**	.53–.71	.62–.74	.78–.91	704.46***	118	.10	.88	.86	228.54***
PS	—									
BDI, BHS	.41**	.48–.67	.64–.73	.78–.91	508.34***	117	.08	.92	.90	196.12***
BHS, PS	.67**									
BDI, PS	—									
BDI, BHS	.70**	.53–.71	.61–.74	.78–.91	315.57***	116	.06	.96	.95	192.77***
BHS, PS	.61**									
BDI, PS	.78**									

Note: Dash (—) indicates uncorrelated items. BDI = Beck Depression Inventory; BHS = Beck Hopelessness Scale; PS = Psychache Scale; RMSEA = root mean square error of approximation; CFI = comparative fit index; TLI = Tucker–Lewis index.
$^{**}p < .01$. $^{***}p < .001$.

variable. Consequently, 31 individuals with missing data were excluded, and a total sample of 556 participants was utilized in the confirmatory factor analyses.

Guidelines suggest acceptable fit when the root mean square error of approximation is .08 or less (Browne & Cudeck, 1993) and when the comparative fit index and Tucker–Lewis index are .95 or greater (Fabrigar, 2007; Hu & Bentler, 1999). The latter indices represent complementary approaches to model fit and so have strong empirical support (Fabrigar, 2007; Kline, 1998). Table 3 displays the correlations among latent variables, factor loadings, and fit indices for each model. A model with three correlated factors outperformed models with zero, one, and two correlated factors. This model—in which the BDI, BHS, and Psychache Scale are correlated with each other—produced fit indices as follows: $\chi^2(116) = 330.18$, $p < .001$; root mean square error of approximation = .06 (90% confidence intervals [CI] = .05 to .07); comparative fit index = .97; and Tucker–Lewis index = .96. In this solution, factors comprising parcels of BDI and BHS items correlated .74 (90% CI = .68, .80); factors comprising parcels of BHS and Psychache Scale items correlated .66 (90% CI = .60, .72); and factors comprising

Table 4
Unrotated Canonical Correlation Analyses

Dimensions	Parcels of Original Items	Parcels of Dichotomous Items
Depression and hopelessness		
Depression		
Variance	.60	.40
Redundancy	.29	.14
Hopelessness		
Variance	.53	.41
Redundancy	.25	.15
R_C	.70	.60
$\chi^2(42)$	405.63	258.77
Hopelessness and psychache		
Hopelessness		
Variance	.51	.52
Redundancy	.20	.19
Psychache		
Variance	.85	.78
Redundancy	.34	.29
R_C	.64	.61
$\chi^2(24)$	300.82	281.52
Depression and psychache		
Depression		
Variance	.61	.42
Redundancy	.35	.17
Psychache		
Variance	.87	.75
Redundancy	.51	.31
R_C	.76	.64
$\chi^2(28)$	519.71	280.80

Note: All chi-square values significant at the $p < .001$ level.

parcels of BDI and Psychache Scale items correlated .81 (90% CI = .78, .84). None of the other three models demonstrated good fit. Because the models were nested, it was possible to determine that each subsequent model had a significantly better fit to the data—that is, a difference in chi-square ($df = 1$) exceeding 3.84—relative to each preceding model consisting of one fewer pair of correlated factors.

Similar results emerged for parcels of dichotomous data (see Table 3). A model with fully intercorrelated BDI, BHS, and Psychache items, produced fit indices of $\chi^2(116) = 315.57$, $p < .001$; root mean square error of approximation = .06 (90% CI = .05, .06); comparative fit index = .96; and Tucker–Lewis index = .95. In this solution, factors comprising parcels of dichotomous BDI and BHS items correlated, .70 (90% CI = .63, .76), factors comprising parcels of dichotomous BHS and Psychache items correlated, .61 (90% CI = .56, .67), and factors comprising parcels of dichotomous BDI and Psychache items correlated, .78 (90% CI = .74, .82). None of the other three models consistently met criteria for good fit. Again, each subsequent model offered a significantly better fit to the data relative to each preceding model with one fewer pair of correlated factors.

Canonical Correlation Analysis

Redundancy estimates from canonical correlation analyses were used to gain information about potential asymmetrical overlap between pairs of constructs (see Table 4). For parcels of original items, only the first canonical correlation was significant in each analysis. When parcels of depression items were related to parcels of hopelessness items, the canonical correlation (R_C) was .70. In this variate pair, depression accounted for 29% of the variance in hopelessness, whereas hopelessness accounted for 25% of the variance in depression. When parcels of hopelessness items were related to parcels of psychache items, the R_C was .64. Here, hopelessness accounted for 20% of the variance in psychache, and psychache accounted for 34% of the variance in hopelessness. Finally, when parcels of depression items were related with parcels of psychache items, $R_C = .76$. Depression accounted for

Table 5
Standardized Regression Weights for Statistically Predicting Suicidal Criteria With Bootstrapping (10,000 Resamples)

		Statistical Predictor[a]		
Criterion	Multiple R^2	Depression	Hopelessness	Psychache
Parcels of original items				
Number of attempts	.10***	.20	–.02	.15
RASQ				
Internal perturbation-based reasons	.34***	.16*	.07	.39***
Extrapunitive/manipulative motivations	.16***	.11	.02	.29***
BSS				
Motivation	.38***	.18	.14*	.35***
Preparation	.23***	.15	.04	.33***
Parcels of dichotomous items				
Number of attempts	.06***	.05	.11	.10
RASQ				
Internal perturbation-based reasons	.31***	.15*	.13*	.33***
Extrapunitive/manipulative motivations	.16***	.16*	.01	.26**
BSS				
Motivation	.29***	.09	.28***	.23**
Preparation	.18***	.07	.14*	.25***

Note: RASQ = Reasons for Attempting Suicide Questionnaire; BSS = Beck Scale for Suicide Ideation.
a. Statistical predictors are factor scores.
*$p < .05$; **$p < .01$; ***$p < .001$.

35% of the variance in psychache, and psychache accounted for 51% of the variance in depression. Patterns of results were similar for the analysis of parcels of dichotomized items, with smaller coefficients likely resulting from range restriction.

Regression Analysis

Factor scores were derived from parcels of items using maximum likelihood extraction and promax rotation of three factors (depression, hopelessness, and psychache). Each suicide criterion was separately and simultaneously regressed on factor scale scores for depression, hopelessness, and psychache in AMOS 6.0 (Arbuckle, 2005), to determine the unique contributions of the latter constructs (see Table 5).

The pattern of results for parcels of original and dichotomous items was comparable such that all regression weights fell within the 90% confidence interval of the other set. For example, the standardized association between hopelessness and internal perturbations ($\beta = .13$; 90% CI = .04, .21) was statistically significant for parcels of dichotomous items but not for parcels of original items ($\beta = .07$); however, the latter weight falls within the 90% confidence interval corresponding to the analysis for parcels of dichotomized items. Psychache was consistently associated with the greatest number of suicide criteria, and it generally accounted for a greater proportion of unique variance in the suicide criteria than that of both hopelessness and depression. Nevertheless, all three variables made unique contributions to the statistical prediction of suicide risk.

Discussion

Suicide is an urgent and potentially preventable problem. Clarifying the relationship between models of depression (Beck, 1967), hopelessness (Abramson et al., 1989), and psychache (Shneidman, 1993) and the kinds of suicidal thoughts and behaviors with which they relate is a crucial step toward the development of a more integrated model for understanding suicide. The emergence of three factors as predicted suggests that depression, hopelessness, and psychache are empirically distinct but highly correlated. The only model that met criteria for good fit was the one in which all three constructs were consistently correlated. Because the findings converged using a variety of heuristics and scoring methods (i.e., parcels of original and dichotomous items), the resulting three-factor structure is less likely to be attributable to differing response formats on the BDI, BHS, and Psychache Scale.

In the present study, communalities exceeding one for data sets using original items and dichotomous items necessitated the use of parceling in subsequent analyses. This was unexpected but not surprising given the mathematical advantages of using item parcels for achieving better model fit—for instance, greater reliability than that of individual items, higher communalities, distributions that more closely approximate normality and interval scaling, a higher indicator to sample size ratio, and less item-idiosyncratic influence (Meade & Kroustalis, 2006). Although Meade and Kroustalis (2006) have cautioned that item parcels are less effective than individual items for detecting differences between conditions—such as the stability of measurement across time, populations, or raters—this was less of a concern with sample sizes exceeding 500, at which point the power to detect measurement invariance was substantially improved.

Research pertaining to key statistical predictors of suicidality has typically involved factor analyses of single inventories with equivocal results. Researchers have made efforts to investigate whether hopelessness is a subset of depression, by conducting factor analysis solely with the BDI. Joiner et al. (2001) found that a subset of BDI symptoms formed a hopelessness dimension distinct from major depression in 1,680 psychiatric outpatients and 1,404 air force cadets. However, this finding has not been replicated (Haslam & Beck, 1994). Others have explored this issue by conducting factor analysis solely with the BHS. Although the original authors reported three factors—namely, Feelings About the Future, Loss of Motivation, and Future Expectations—it is not clear whether such a structure supports Beck's model of depression (1967) or Abramson et al.'s model of hopelessness depression (1989). Between three and five BHS factors have been reported elsewhere (Holden, 1986). Thus, in the present study, the inclusion of items from measures designed to tap all variables of interest is a novel approach and one that clarifies the state of current knowledge regarding the relationship between key statistical predictors of suicide.

The overlap between depression, hopelessness, and psychache was asymmetrical as hypothesized. Interestingly, psychache accounted for about half the variance in depression and a third in hopelessness. The data fail to support Abramson and colleagues' (1989) model. If hopelessness was a subtype of depression, one would have expected depression to account for 100% of hopelessness, which was not the case. Results clearly indicate that one can be hopeless without being depressed. The possibility of only partial overlap with psychache also diverges from Shneidman's (1993) view that psychache fully mediates the relationship between all suicide-relevant variables and suicide (i.e., no psychache, no suicide). Given the present findings, we propose an integrative model in which (a) psychache overlaps to a large extent with depression and a moderate extent with hopelessness, (b) depression and hopelessness overlap to a similar and moderate degree with each other, and (c) it is possible to have high scores on only one or two of these constructs, but (d) the individuals at highest risk for suicide are those with the highest levels of psychache.

When suicide criteria were simultaneously yet separately regressed on each statistical predictor, psychache was consistently associated with the greatest number of suicide indices—namely, internal perturbation–based reasons, extrapunitive/manipulative reasons, the motivation to commit suicide, and one's preparedness to undertake suicidal action. Depression and hopelessness also emerged as unique statistical suicide predictors, but the strength of their relationship with the criteria varied—depending on whether parcels of original or dichotomous items were used—and their standardized associations with suicide criteria were often only half as strong as that of psychache. Thus, psychache was the component most consistently and clearly indicative of proneness toward self-destruction.

The present findings generally support the claim by Shneidman (1993) that psychache is the primary reason that people kill themselves. Although Shneidman first coined the term *psychache* to refer to the unbearable psychological pain that arises from frustrated psychological needs, the broader concept of mental pain has been viewed as the root of psychopathology in psychiatry since the early 20th century. From a psychodynamic view, mental pain has been defined as the feelings of mourning and longing for a loved one that occurs following a traumatic loss (Freud, 1917/1955). Existentialists and emotion-focused therapists have viewed mental pain as arising primarily from a sense of emptiness and meaninglessness (Frankl, 1963), as a disruption of one's sense of wholeness (Bakan, 1968), or as a sense of self-brokenness (Bolger, 1999). More recent cognitive theorists have posited that mental pain is an extreme form of disappointment in oneself that is caused by a discrepancy between one's actual and ideal views of oneself (Baumeister, 1990). By redefining mental pain as psychache, Shneidman has created an operationalization of this construct that facilitates scientific inquiry in suicidology and that transcends any theoretical point of view.

Researchers have demonstrated that psychological factors mediate the effects of nonpsychological variables on suicide risk. For instance, Abramson et al. (1989) showed that hopelessness accounts for the relationship between suicidality and variables such as poor problem solving, parental depression, and personality disorder. In addition, Berlim et al. (2003) found that poor psychological quality of life—as used as a proxy for psychache—can eliminate the relationship among suicidality and physical, interpersonal, and environmental quality of life, even when depression and hopelessness are controlled. In our integrative model of suicide risk, we acknowledge the potential role of environmental stressors, although the means by which they interact with psychological variables awaits clarification through further research.

Limitations of the present study must be also considered. First, a predominantly female university sample was used, which may limit the generalizability of these results. However, there were no significant gender or age differences on any of the variables of interest in the present study. As well, findings based on analogue and clinical samples in depression and attribution research have generally been consistent (Vredenburg, Flett, & Krames, 1993). A focus on young adults is relevant given the emphasis on early intervention and suicide prevention, as well as Shneidman's (1971) finding that prodromal life patterns associated with eventual suicide completion are set by age 30. Evidence that psychache is associated with suicide criteria in male prison inmates and psychiatric outpatients, even when hopelessness and depression are controlled (Berlim et al., 2003; Holden & Kroner, 2003), suggests that the present findings may have relevance in high-risk populations as well as college-age populations. Given suggestions that prison inmates have difficulty distinguishing negative affect dimensions (e.g., anxiety, depression) from cognitive dimensions (e.g., persecutory ideas; Mills et al., 2005), the relationships between depression, hopelessness, and psychache in inmates may resemble those found in university students, for whom differentiation between constructs may have been limited by range restriction in the scores obtained. However, further research is needed to ascertain how depression, hopelessness, and psychache overlap in a wider range of populations.

Second, only self-report instruments were used to measure depression, hopelessness, and psychache. Although concerns about the appropriateness of self-report over structured clinical interviews have been expressed as part of the analogue–clinical debate, studies of suicide risk using these two methods have led to the same conclusions regarding the relationship of constructs to each other. Abramson et al. (1989) cautiously advocate the use of self-report instruments, given their practical advantages. However, replication is needed, using alternative measures of depression, hopelessness, and psychache in a wider range of samples, to confirm the generalizability of present findings.

In particular, analyzing existing scales as parcels of items only partially addresses concerns regarding the comparability of the BDI, BHS, and Psychache Scale. The mixture of low- and high-frequency items on a particular questionnaire, as well as the period of reference used, may influence interpretations of items and subsequent scale ratings. For instance, Schwarz (2003) found that participants tend to report more severe events when they are asked how frequently they experienced an event over the past year than when they are asked to rate their experience of an event over the past week. Furthermore, Tourangeau and Smith (1996) found that when individuals decide how often they experience an event, they tend to rate that event in comparison to the frequency of the other events on the questionnaire.

For example, respondents may rate an event (e.g., feeling sad) as occurring infrequently when the other test items are relatively common (e.g., feeling tired, experiencing changes in sleep or eating habits); conversely, they may rate the same item (feeling sad) as occurring frequently when most of the other items on a questionnaire are relatively rare (e.g., thoughts of killing oneself). In the present study, several issues associated with scale format may have influenced participants' ratings. The BDI and BHS ask participants to consider the frequency of various experiences over the past week, including the present day. Both these questionnaires include a mixture of low- and high-frequency events. In contrast, the Psychache Scale contains relatively rare events and does not specify a reference period. Despite these issues, one may speculate that ratings of highly salient, concrete, and rare experiences, such as suicidal behavior, are less prone to shifts in endorsement frequency simply because of variations in scale format. The floor effects obtained on the BDI, BHS and Psychache Scale support this claim. Whether a test uses a true/false format (i.e., BHS) or a point-based scale (BDI: 0–3; Psychache Scale: 1–5) and regardless of the other items contained or the reference period specified, suicidality is endorsed relatively infrequently. It is this more serious category of events that is of interest in the study of suicide risk. Using the original scales in the present study also allows findings to be compared with previous research. Employing equivalent scale formats to assess depression, hopelessness, and psychache would not have addressed the possibility that altering response formats may also alter the meanings that participants derive from questionnaire items. Nevertheless, the clear pattern of results in the present study, highlighting the prominence of psychache, suggests that our findings are not likely attributable to method variance.

Finally, key psychological predictors of suicidality were studied in relation to proxies for suicide completion. Only 18% to 38% of suicide attempts lead to intentional death despite the fact that a history of suicide attempts increases one's risk by at least fivefold (Clark & Fawcett, 1992). Nevertheless, it is a logical strategy to target the thoughts and painful feelings that precede self-destructive acts, as a means of ultimately preventing suicidal behavior.

In summary, our results corroborate other work suggesting that psychache is the preeminent statistical predictor of suicide risk (Berlim et al., 2003; DeLisle & Holden, 2004; Flamenbaum & Holden, 2007; Holden et al., 2001). We have presented an integrative model that clarifies for the first time how psychache fits in relation to other psychological risk predictors, such as depression and hopelessness, with opportunities for future exploration of the role that additional variables play. The general pattern of findings was consistent when item format varied. Given that psychache is distinct from hopelessness and depression and that it more strongly relates to important suicide criteria than purely cognitively derived risk predictors, the integration of psychache into broader, cross-theoretical models of suicide risk is warranted.

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Michelle M. DeLisle, PhD, is currently a Registered Provisional Psychologist at the Misericordia Hospital in Edmonton, Albera, Canada. Her research interests include suicide prevention, mental health and wellness, and program evaluation. Previous publications in these areas include 8 journal articles and 2 book chapters.

Ronald R. Holden is a Professor of Psychology at Queen's University, Kingston, Ontario, Canada. He received his undergraduate degree from the University of Toronto before completing his MA and PhD degrees in Clinical and Experimental Psychopathology at the University of Western Ontrario. Dr. Holden conducts research and publishes on the topics of test construction, psychometrics, and suicide.